CTS SRE Day 4

Performance Metrics with SLI/SLO

EMPOWERING High Performance Technology Teams

Agenda for the Day

Welcome and Objectives (5 minutes)
Understanding the importance of performance metrics for reliability.

SLAs, SLOs, and SLIs: Concepts and Hierarchies (15 minutes)
Defining and understanding their relationships for service reliability.

Selecting Metrics and Setting Targets (15 minutes)
Aligning with business objectives and ensuring system health.

Error Budget Calculation and Monitoring (15 minutes)
Managing reliability thresholds and proactive monitoring.

Case Study: SLI/SLO Deployment (20 minutes)
Real-world application for authentication and payment systems.

Interactive Exercises and Q&A (10 minutes)
Applying concepts to scenarios and open discussions.

Note: For sessions that are completely demo/Lab or notebook based, please keep one agenda slide and explain the same thing in the beginning so that participants are aware they do not have PPT.

SLIs, SLOs, and SLAs: Measuring Reliability

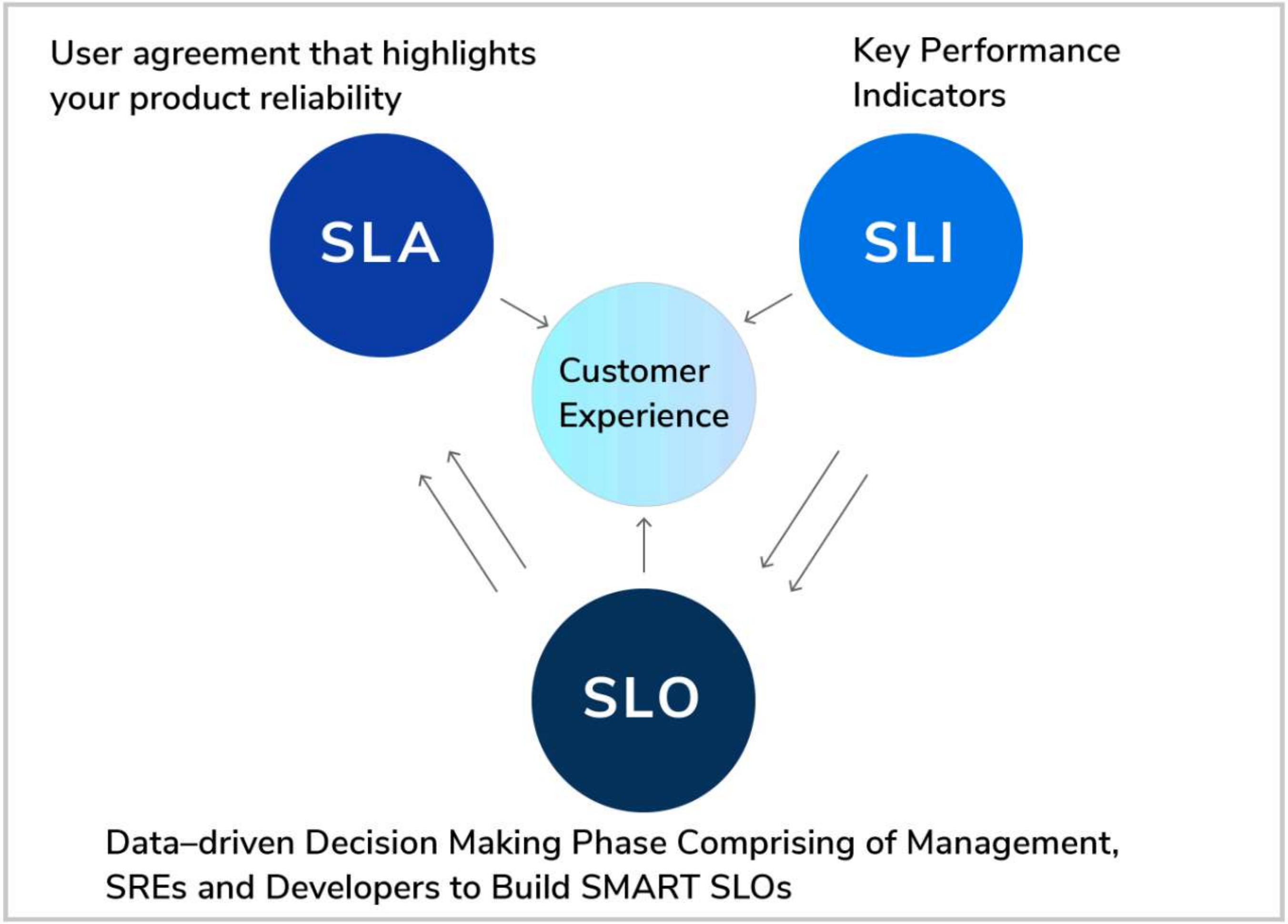
- **SLI (Service Level Indicator):** Metrics that define reliability, such as *latency* or *error rates*.
- **SLO (Service Level Objective):** Target performance for SLIs (e.g., 99.9% availability).
- **SLA (Service Level Agreement):** Formal agreements with customers on SLO expectations.
- **Use Case Example: Google**:
- Google's **YouTube** team uses SLIs for video upload latency and SLOs for availability to ensure minimal downtime for users.

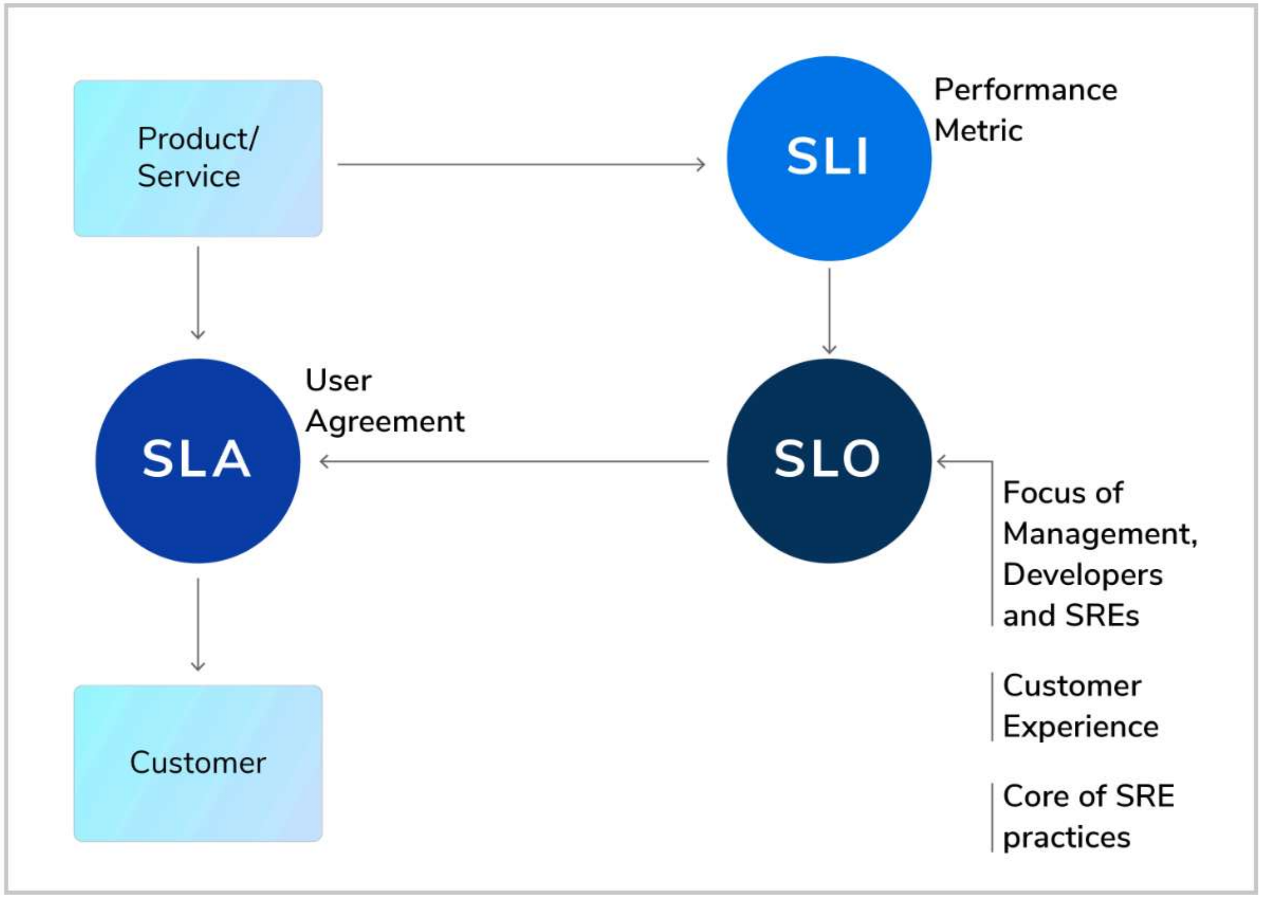
Three Related Elements in SRE

SLI →
- Key Performance Metric
- Measures Product Characteristics
- Calculated in %

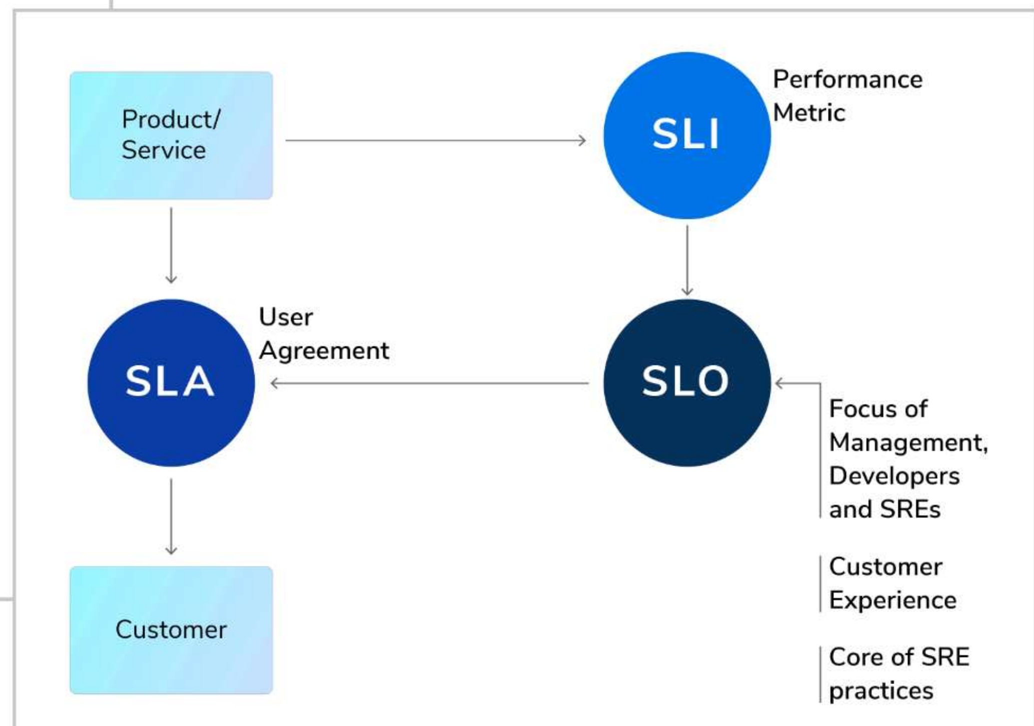
SLO →
- Core of SRE practices
- Data-Driven Approach
- Build SMART SLOs

SLA →
- User Agreement
- Mentions Reliability Values
- If breached, business can incur loss/penalty


User agreement that highlights your product reliability
Key Performance Indicators
SLA
SLI
Customer Experience
SLO
Data–driven Decision Making Phase Comprising of Management, SREs and Developers to Build SMART SLOs


Product/
Service
SLI
Performance
Metric
SLA
User
Agreement
SLO
Focus of
Management,
Developers
and SREs
Customer
Experience
Core of SRE
practices
Customer

Three Related Elements in SRE

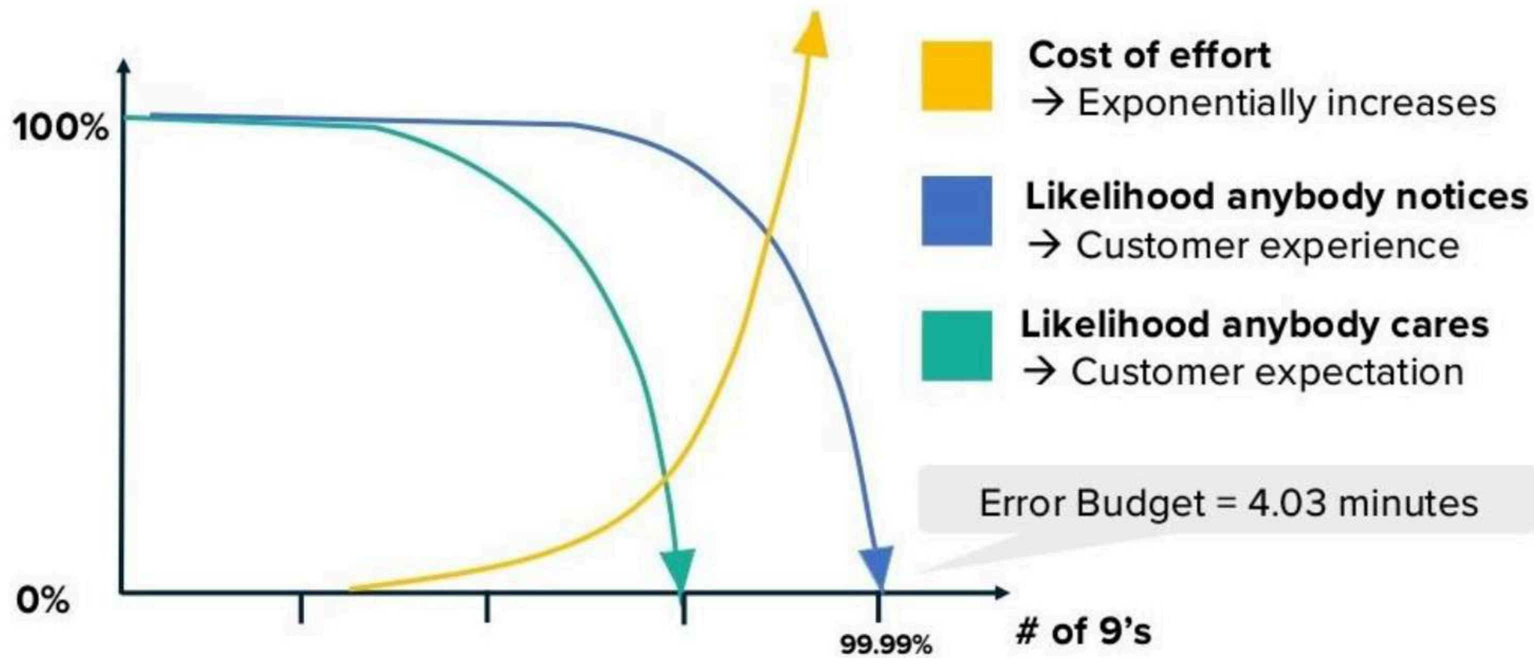


SRE – Tenets & Principles

1. *Know the Service Level*
2. *Embrace Risk*
3. *Eliminate Toil*
4. *Know What's Broken and Why?*
5. *Stuff Happens !. ….Murphy is my Brother*
6. *Automate [Almost] Everything*
7. *Reliable Releases*
8. *Keep it Simple and Stupid*

Error Budget

The amount of error that your service can accumulate over a certain period before your users start being unhappy



Error Budget - Calculation

100% Availability is not your target.

So what is?

SLO	**Error Budget** (per 28 Days)
99%	403 mins
99.5%	202 mins
99.9%	40.3 mins
99.95%	20.2 mins
99.99%	4.03 mins
99.999%	0.40 mins

Agree your SLI's and SLO's with everyone.

Yes, everyone.

- Everyone understands the importance of reliability
- Everyone understands the error budget and how it works
- Everyone understands the new rules!

Oops! We broke something.

What now?

- On-call / Playbooks / Fire drills
- Blameless Incident Review / Retrospective
- Review error budget
- Reduce risk and invest in reliability

Measuring Reliability Targets and Monitoring System Health in Distributed Systems

Key Practices:

Choose SLIs that align with customer experience:

- Latency, availability, throughput, and error rates.

Use tools like Prometheus, Grafana, or New Relic to monitor distributed systems.

Focus on the **Four Golden Signals**: Latency, Traffic, Errors, and Saturation.

Example:

For a payment gateway:

- Latency: Time to process a transaction.
- Availability: Percentage of successful transactions.
- Errors: Failed or timed-out transactions.
- **Discussion Prompt**:

What SLIs would you track for a high-traffic e-commerce checkout system?

Setting Performance Targets, and Aligning with Business Objectives

Steps to Select and Set Metrics:

Identify critical user journeys (e.g., search, checkout).

Map SLIs to business goals.

Define SLO thresholds based on acceptable risk and error budgets.

Example SLO:

Authentication Service:

- 99.95% availability.
- 95% of login requests processed in <500ms.
- **Interactive Exercise**:

Define SLIs, SLOs, and SLAs for a subscription-based video streaming platform.

Error Budget Calculation and Monitoring Implementation

Error Budget Basics:

Total time allowed for service unavailability (derived from SLO).

Formula:

- Error Budget = (100% - SLO Target) × Total Time Period.
- **Example Calculation**:

If SLO is 99.9% uptime:

- Total allowable downtime per month = (1 - 0.999) × (30 days × 24 hours) = ~43.2 minutes.
- **Threshold Setting and Monitoring**:

Use tools like Datadog, PagerDuty for alert thresholds.

Automate actions when error budgets are close to exhaustion.

Scenario:

Discuss how to prioritize bug fixes versus feature rollouts when error budgets are consumed.

Breakout

Case Study: SLI/SLO Deployment for Authentication and Payment Processing Systems

Scenario:A payment processing platform serving 1 million daily transactions needs to ensure high reliability during peak hours.

SLI/SLO Design:

Authentication Service:

SLI: Percentage of successful logins.

SLO: 99.95% success rate.

Payment Service:

SLI: Average transaction processing time.

SLO: 95% of transactions completed within 2 seconds.

Think tank

- **What is an example of a measurable SLI for a login service? How would you capture it?**
- **How would you determine which SLI metrics matter most for a payment system?**
- **What factors should you consider when setting an SLO for response time?**
- **How can you ensure the SLIs you capture align with customer expectations?**
- **What tools can you use to monitor SLIs in real time and alert when SLOs are breached?**

Overall Summary

Key Concepts Covered

SLIs, SLOs, and SLAs: Definitions and their role in reliability.

Hierarchical relationships between SLIs, SLOs, and SLAs.

Core Practices

Identifying critical metrics aligned with business goals.

Setting performance targets and managing error budgets.

Implementation Insights

Tools and techniques for monitoring and proactive management.

Real-world case study: Authentication and payment systems.

Key Takeaways

SLI/SLO management drives reliability, scalability, and customer satisfaction.

Practical frameworks for balancing innovation and stability.

Thank You!

StackRoute

(An NIIT Venture)